

# Изкуствен интелект

***Тема 5: Рекурентни и локално  
рекурентни невронни мрежи  
(RNN, DRNN, LRPNN, GLRPNN, pcLRPNN)***



# Съдържание

- ❑ Обща класификация на методите
- ❑ Неврони с последователен вход. Неврони с КИХ и БИХ. Невронни мрежи с време-закъснителни вериги във входа (Time-Delay Neural Networks)
- ❑ Неврони с обратни връзки. Видове обратни връзки.
- ❑ Архитектури на рекурентни (RNN) и локално рекурентни невронни мрежи
  - DRNN,
  - LRPNN,
  - GLRPNN
  - pcLRPNN...

# Класификатори

## *Discriminative approaches*

LDA

Polynomial  
classifier

TDNN  
and RNN

FFNN

SVM

Decision  
trees

## *Non-discriminative approaches*

k-NN

LVQ

SOM

PNN

GMM

HMM

## *Generative approaches*

## *Combined methods*

GMM/SVM

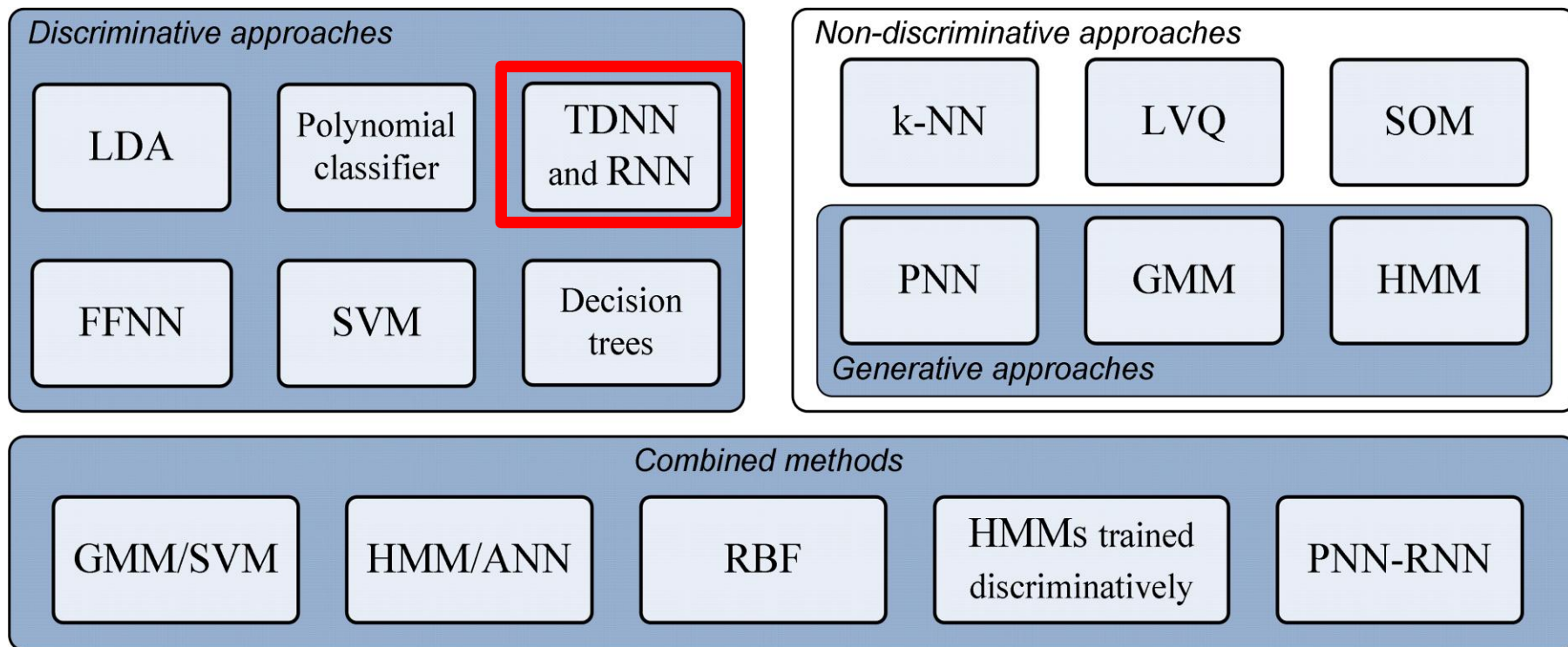
HMM/ANN

RBF

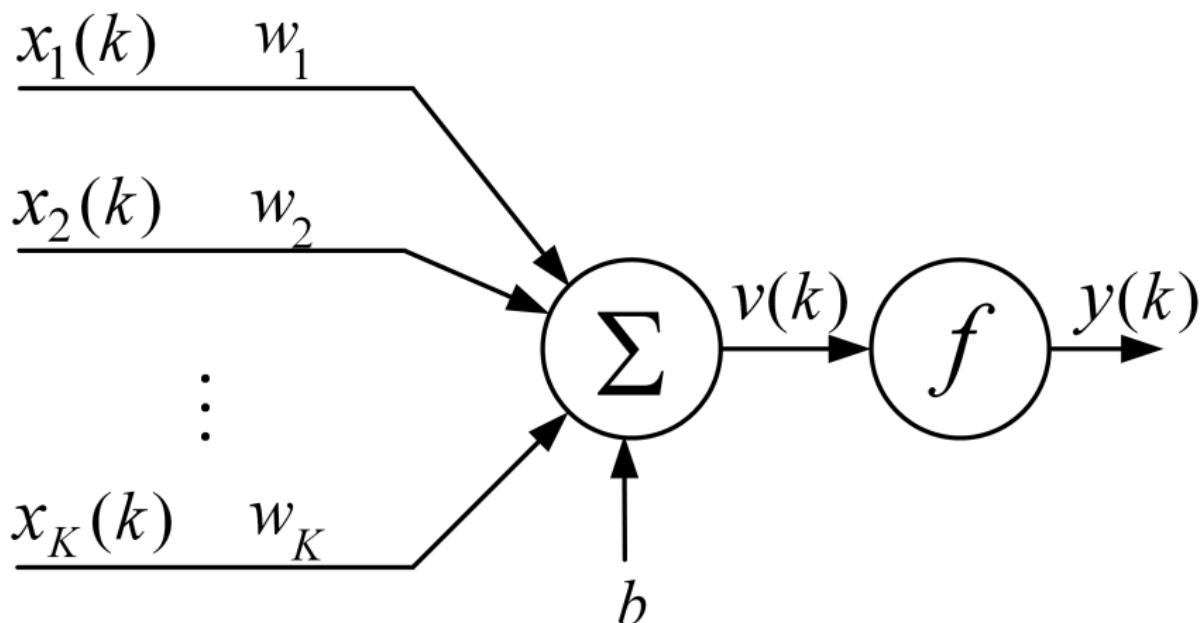
HMMs trained  
discriminatively

PNN-RNN

# Класификатори

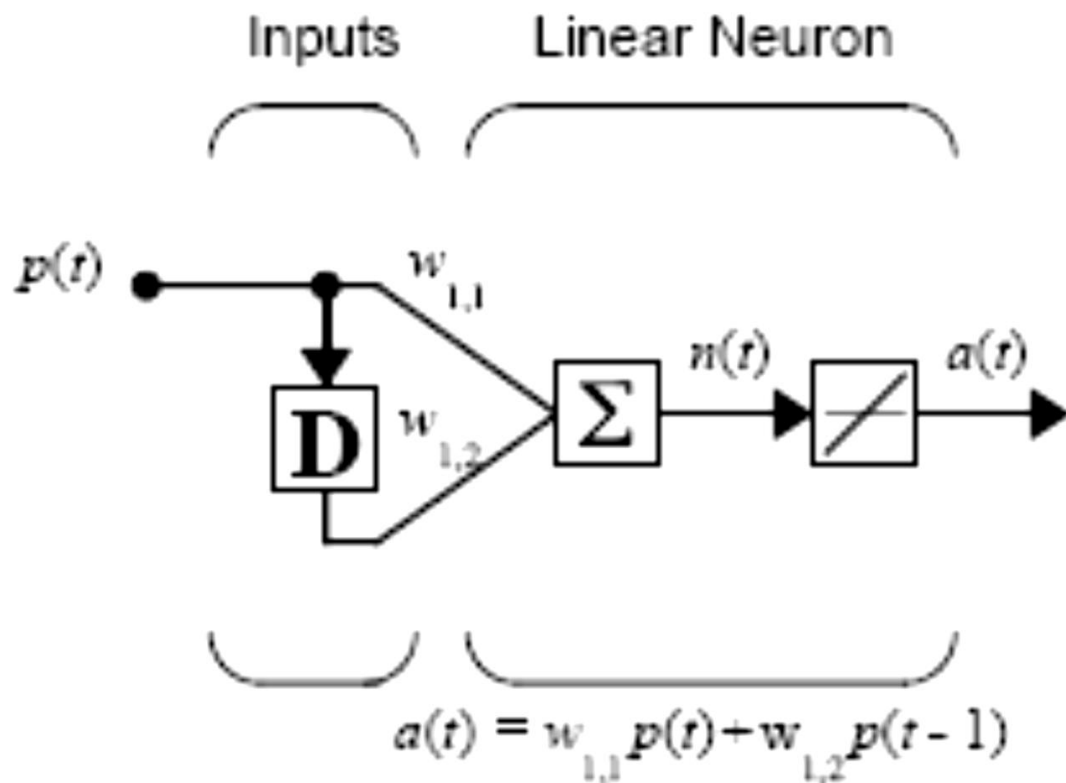


# Неврон без обратна връзка



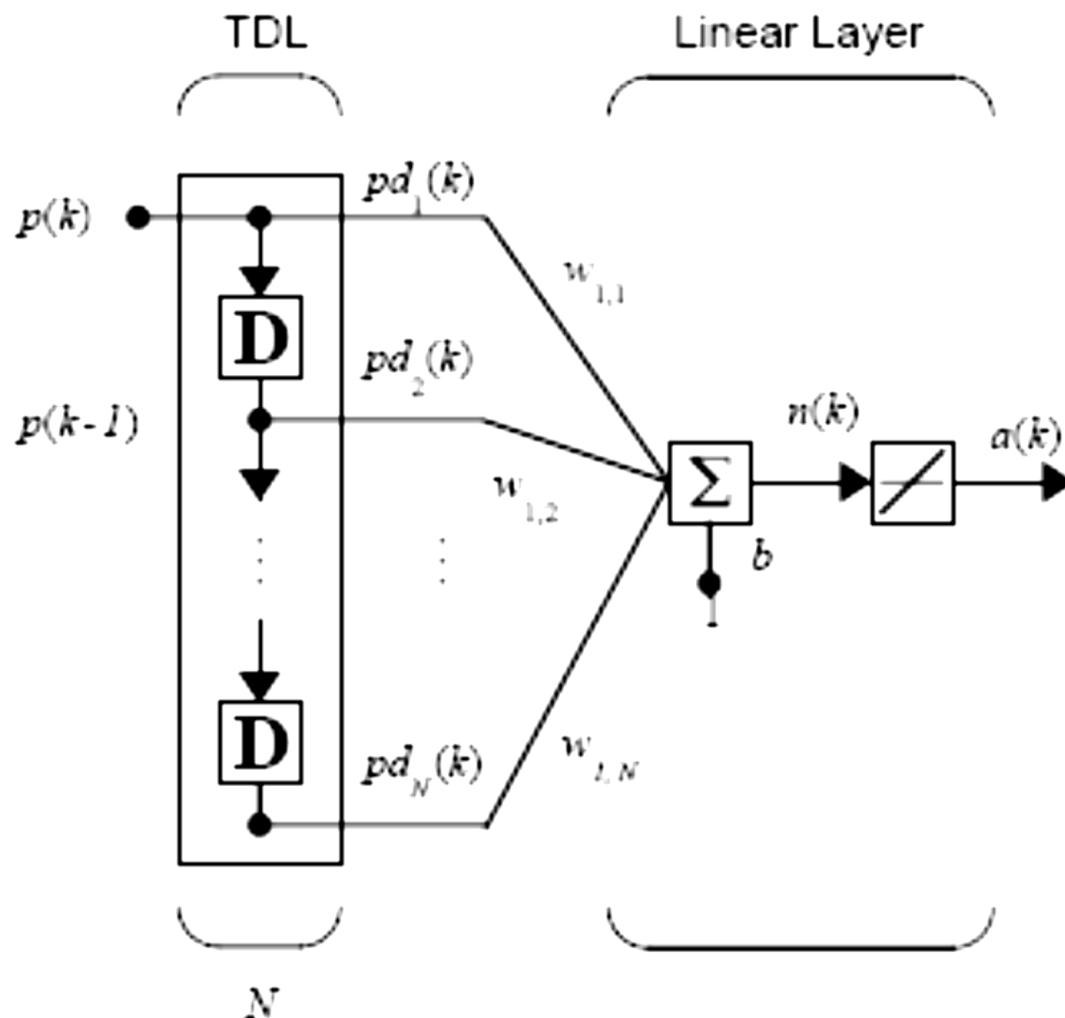
$$y(k) = f(v(k))$$
$$v(k) = \sum_{i=1}^K w_i x_i(k) + b$$

# Неврон с последователен вход

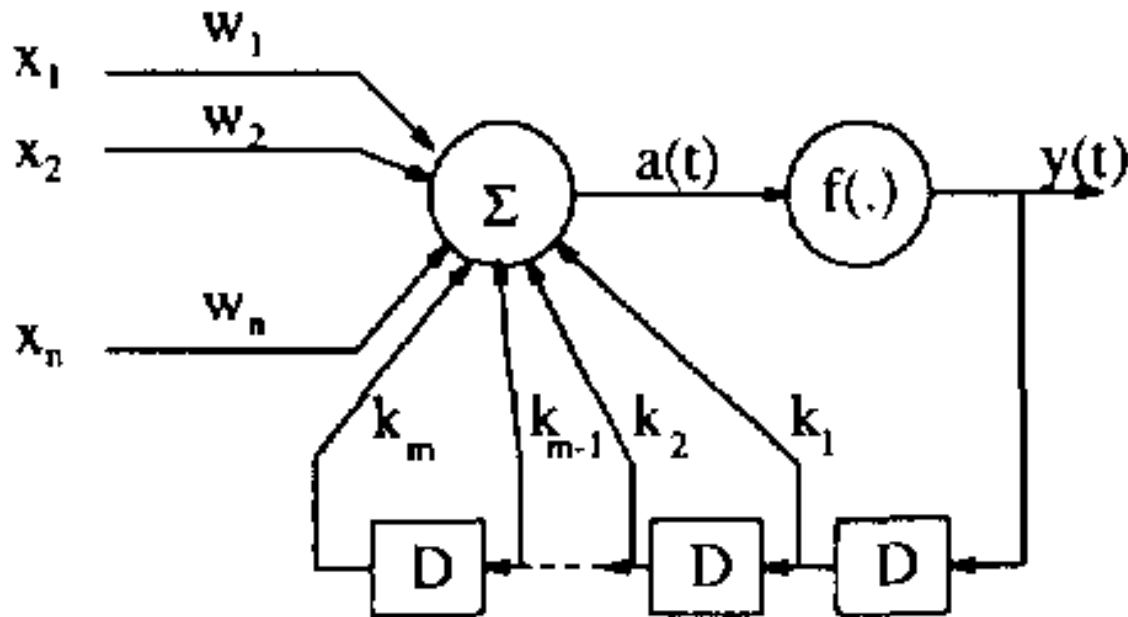


# Неврон с последователен вход

FIR Adaptive Filter

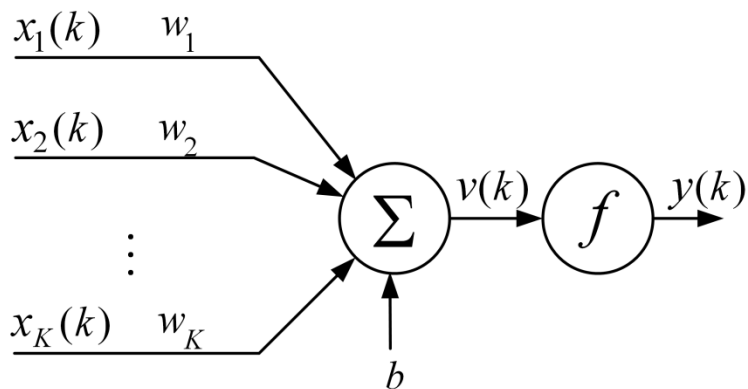


# Неврон с обратни връзки (IIR неврон)

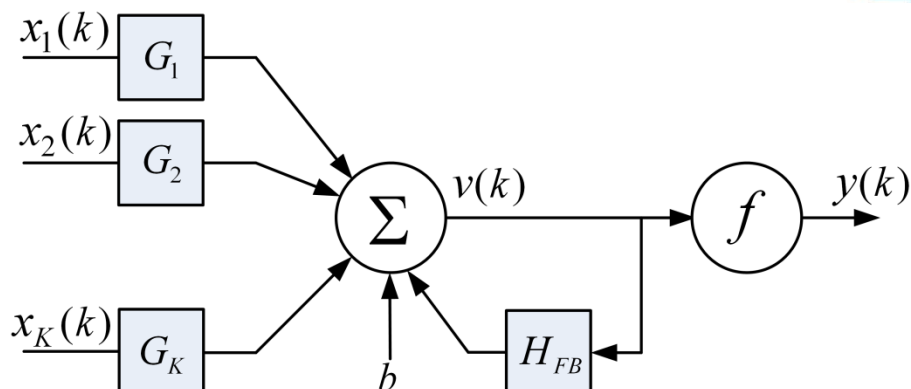




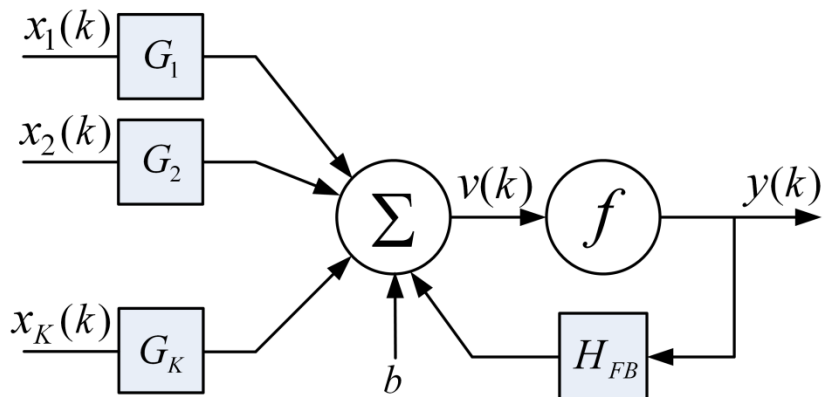
# Локално рекурентни неврони



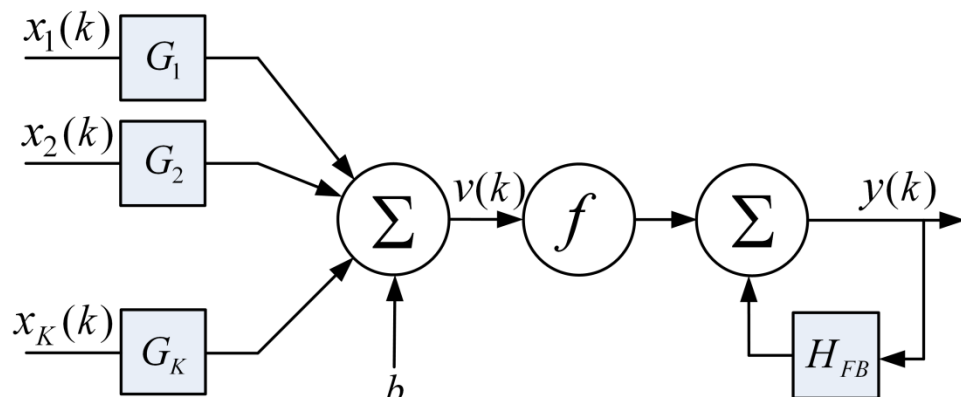
(a)



(b)



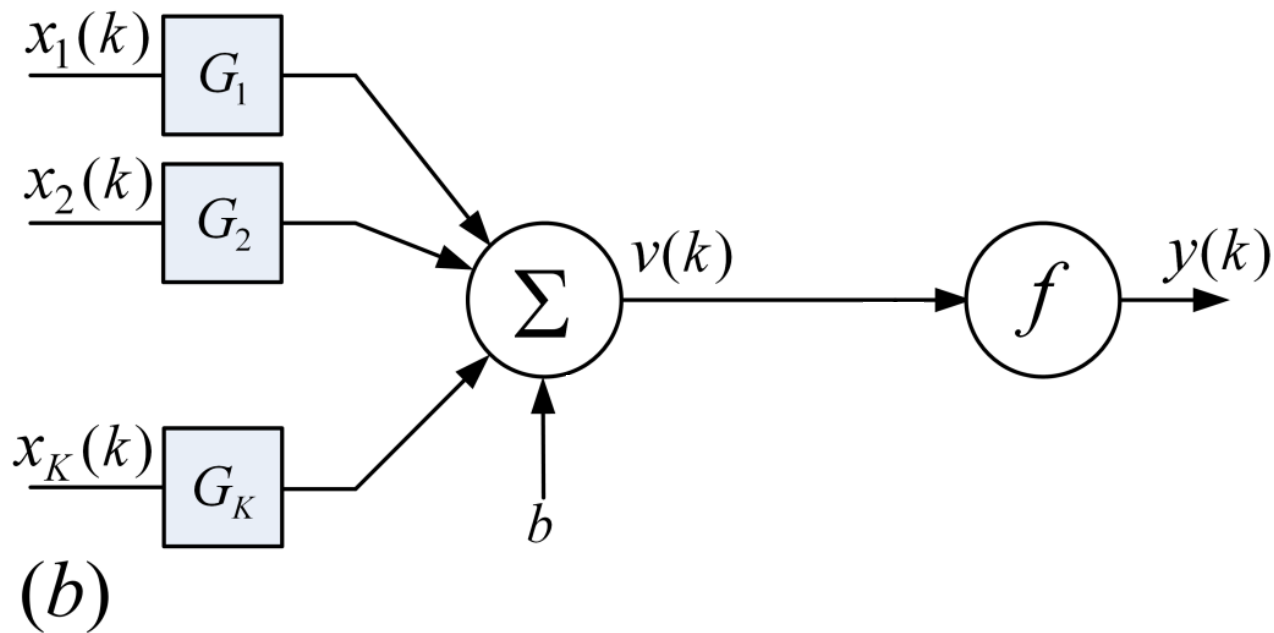
(c)



(d)

- С обратна връзка от изхода на синапса,
- С обратна връзка преди активиращата функция ,
- С обратна връзка от изхода на неврона
- С авторегресивен филтър след изхода на неврона

# Локално рекурентни неврони

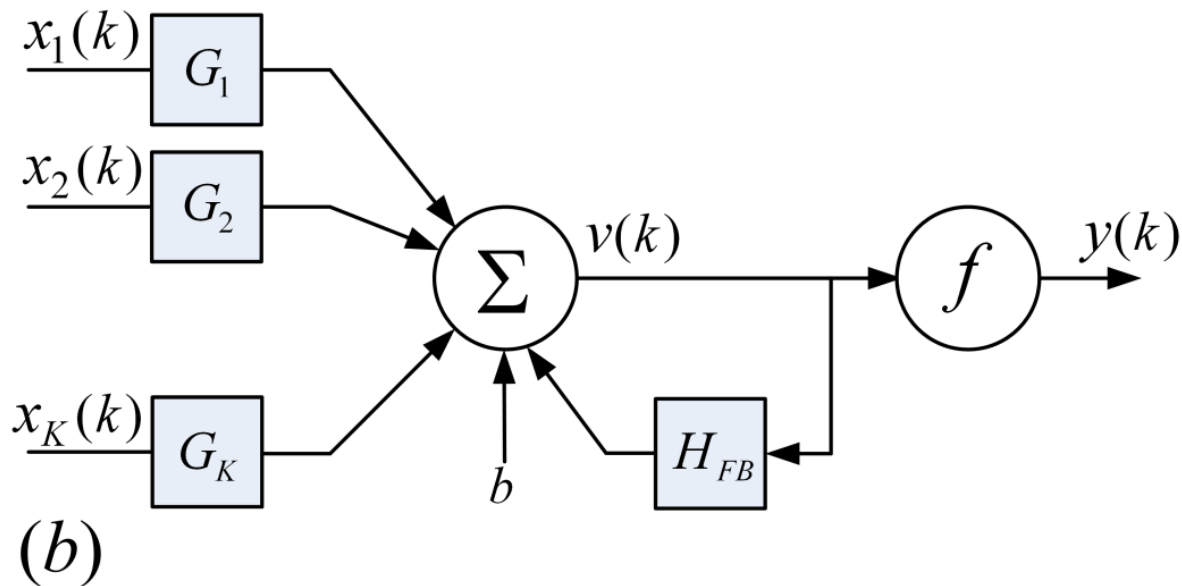


$$y(k) = f(v(k))$$

$$v(k) = \sum_{i=1}^K G_i(z^{-1})x_i(k) + b ,$$

$$G(z^{-1}) = \frac{\sum_{j=0}^{m_z} b_j z^{-j}}{\sum_{j=0}^{n_p} a_j z^{-j}}$$

# Локално рекурентни невронни мрежи

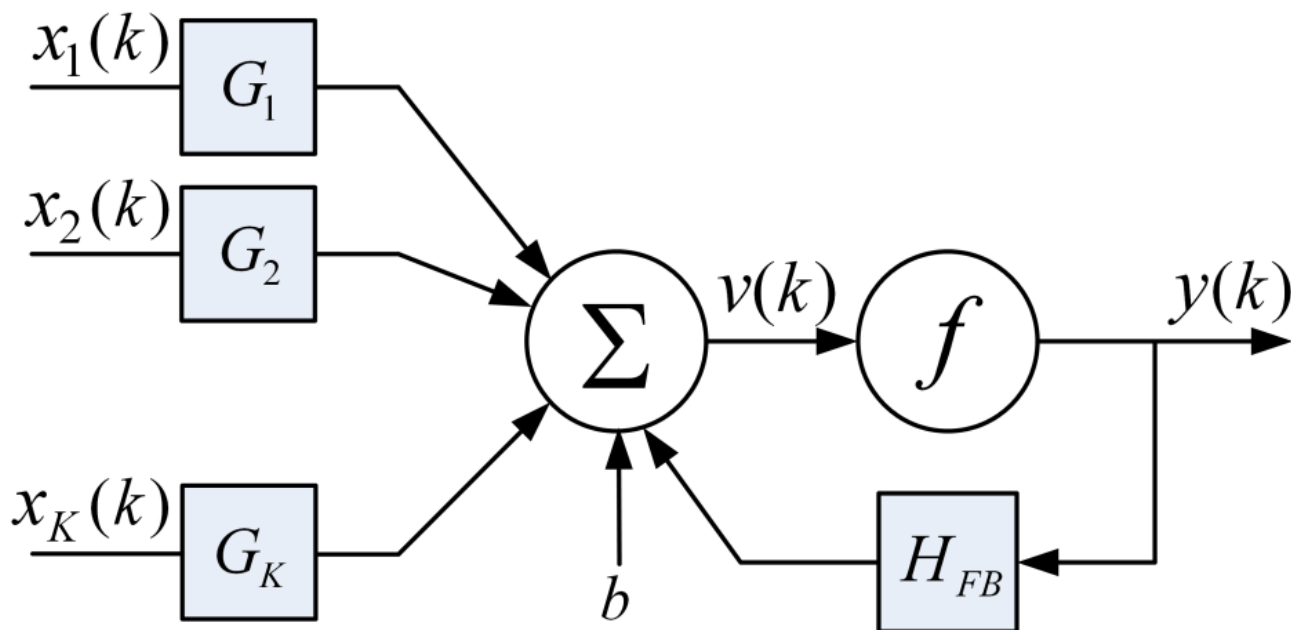


$$y(k) = f(v(k))$$

$$v(k) = \sum_{i=1}^K G_i(z^{-1})x_i(k) + H_{FB}(z^{-1})v(k-1) + b,$$

$$H_{FB}(z^{-1}) = \frac{\sum_{j=0}^{g_z} b_j z^{-j}}{\sum_{j=0}^{l_p} a_j z^{-j}}$$

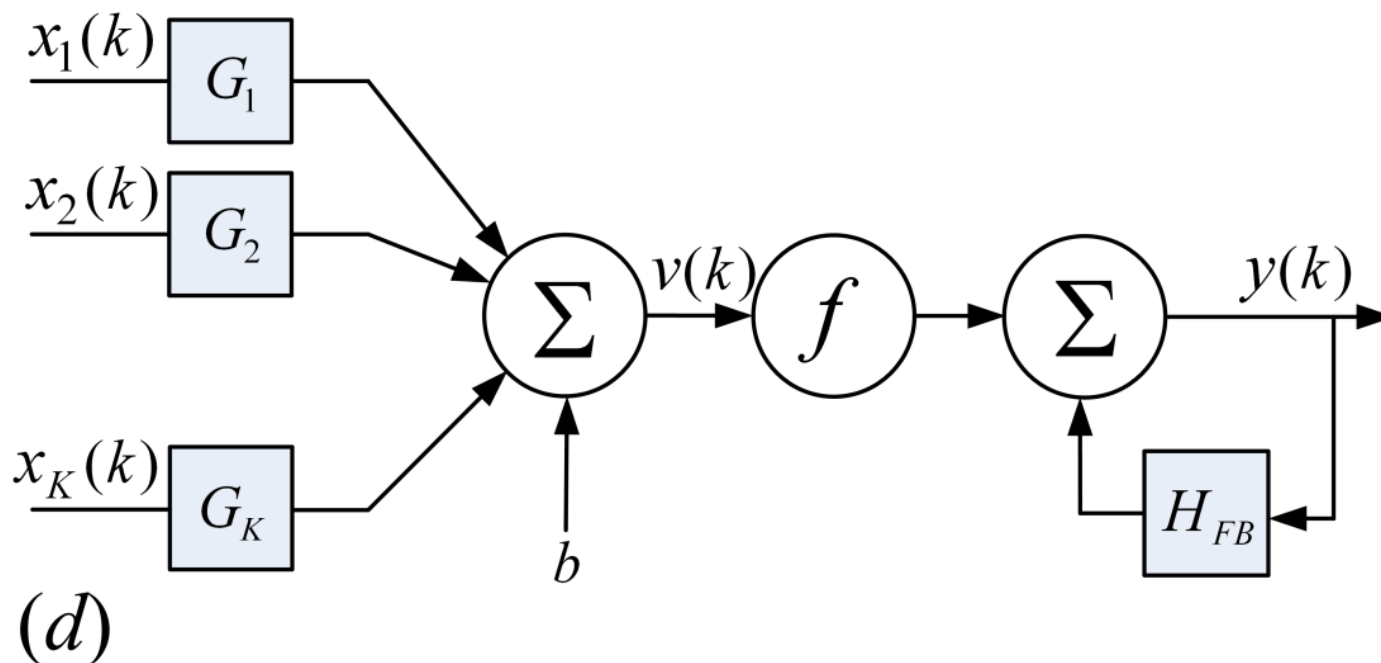
# Локално рекурентни невронни мрежи



$$y(k) = f(v(k))$$

$$v(k) = \sum_{i=1}^K G_i(z^{-1})x_i(k) + H_{FB}(z^{-1})y(k-1) + b .$$

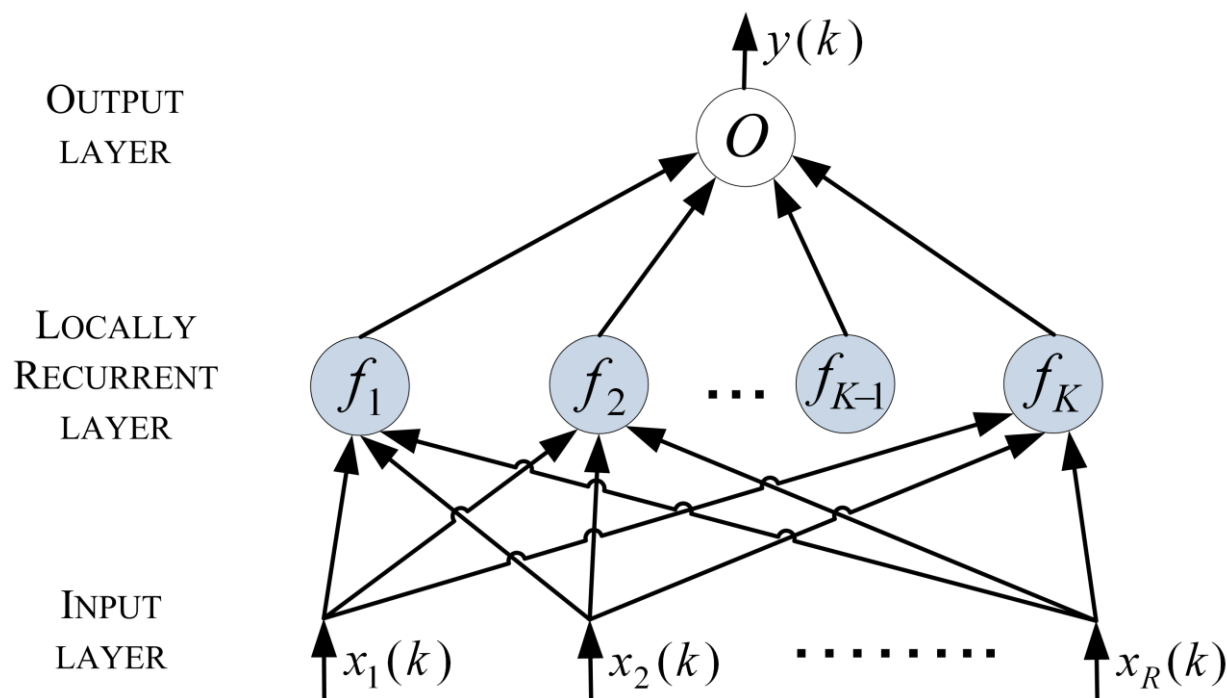
# Локално рекурентни невронни мрежи



$$y(k) = f(v(k)) + H_{FB}(z^{-1})y(k-1)$$

$$v(k) = \sum_{i=1}^K G_i(z^{-1})x_i(k) + b .$$

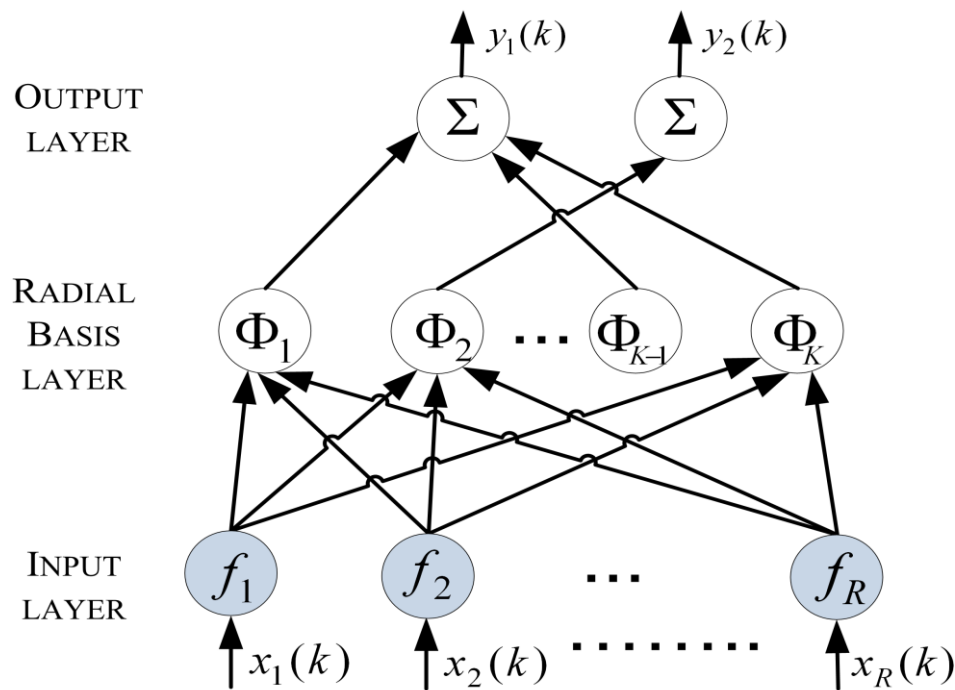
# Диагонално рекурентни невронни мрежи (DRNN)



$$y(k) = \sum_{j=1}^K w_j^O u_j(k)$$

$$u_j(k) = f_j \left( \sum_{i=1}^R w_{ij}^I x_i(k) + w_j^H u_j(k-1) \right),$$

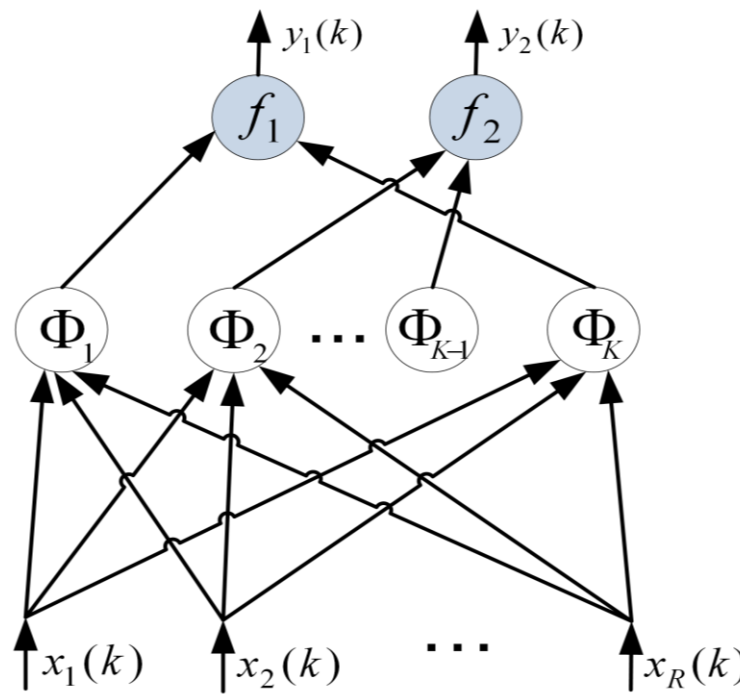
# Locally Recurrent Radial Basis Function Neural Networks



$$y_l(k) = \sum_{j=1}^{K_l} w_j^O \exp\left(-\frac{\sum_{i=1}^R (u_i(k) - \mu_{ij})^2}{\sigma_j^2}\right)$$

$$u_i(k) = f_i(x_i(k) + w_i^I u_i(k-1)) ,$$

# Locally Recurrent Radial Basis Function Neural Networks

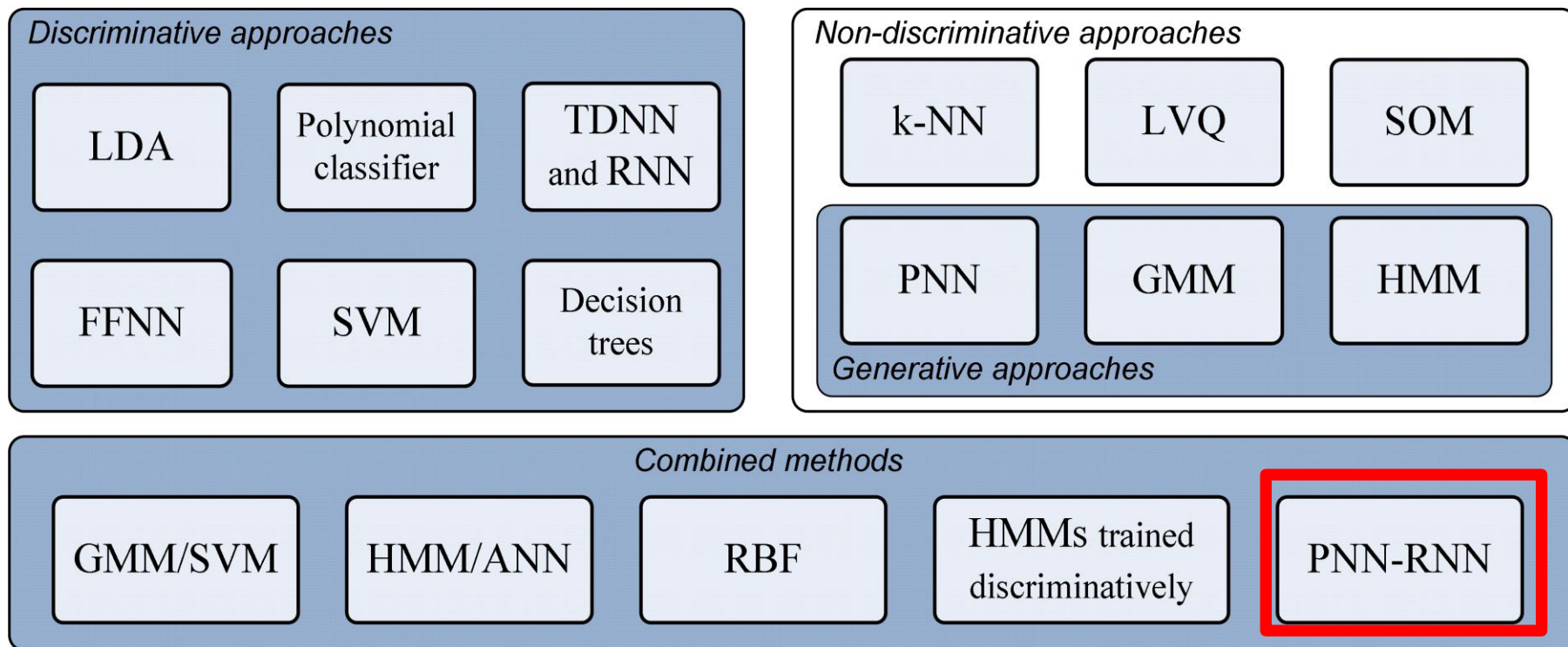


$$y_l(k) = f_l\left(\sum_{j=1}^{K_l} G_j(z^{-1})u_j(k)\right)$$

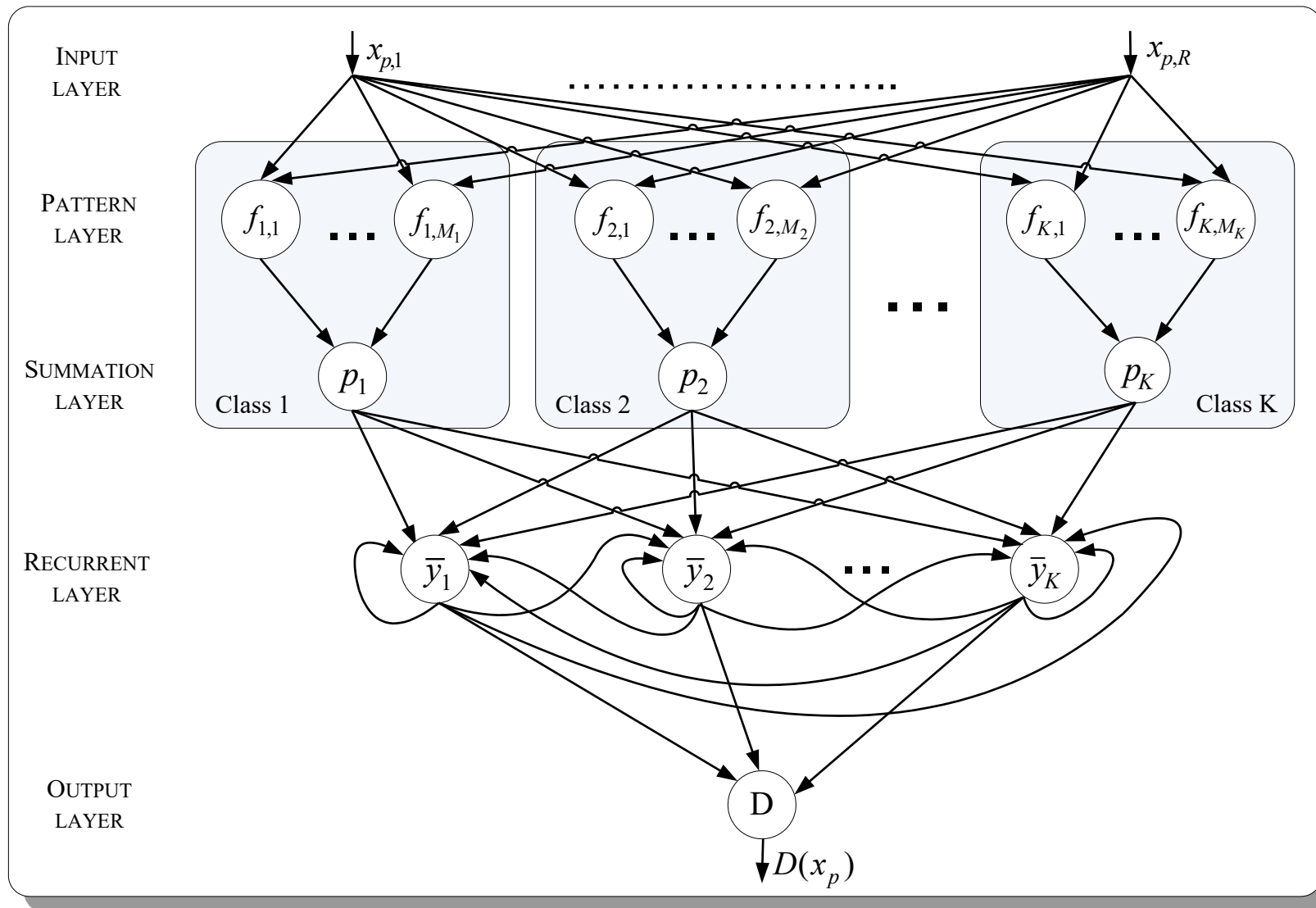
$$u_j(k) = \exp\left(-\frac{\sum_{i=1}^R (x_i(k) - \mu_{ij})^2}{\sigma_j^2}\right),$$



# Класификатори



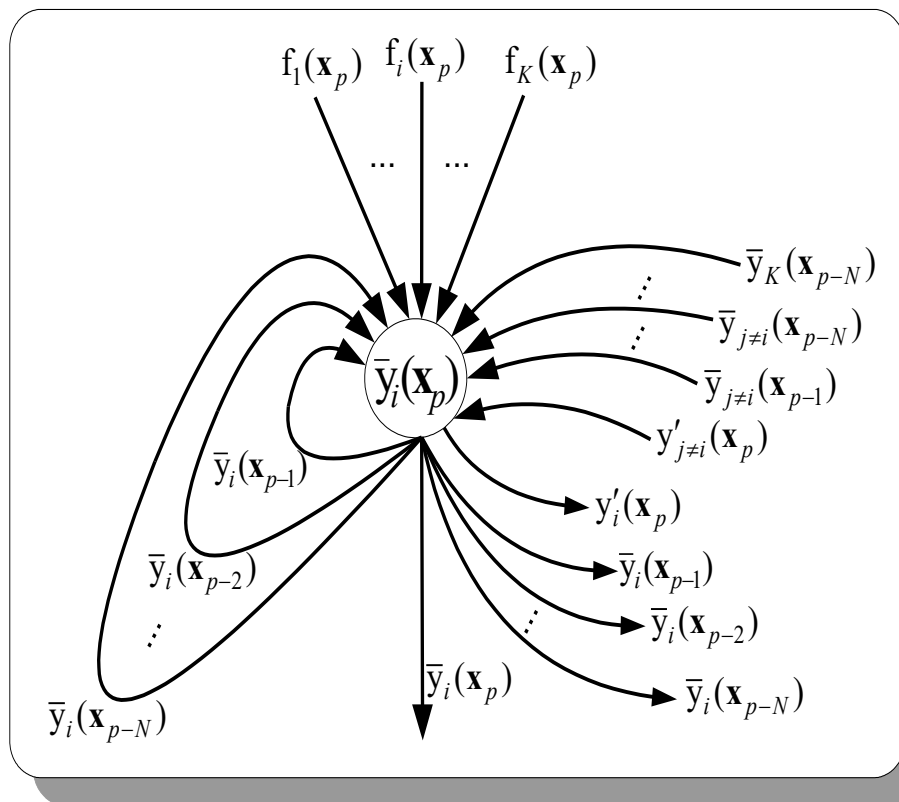
# Locally Recurrent Probabilistic Neural Networks



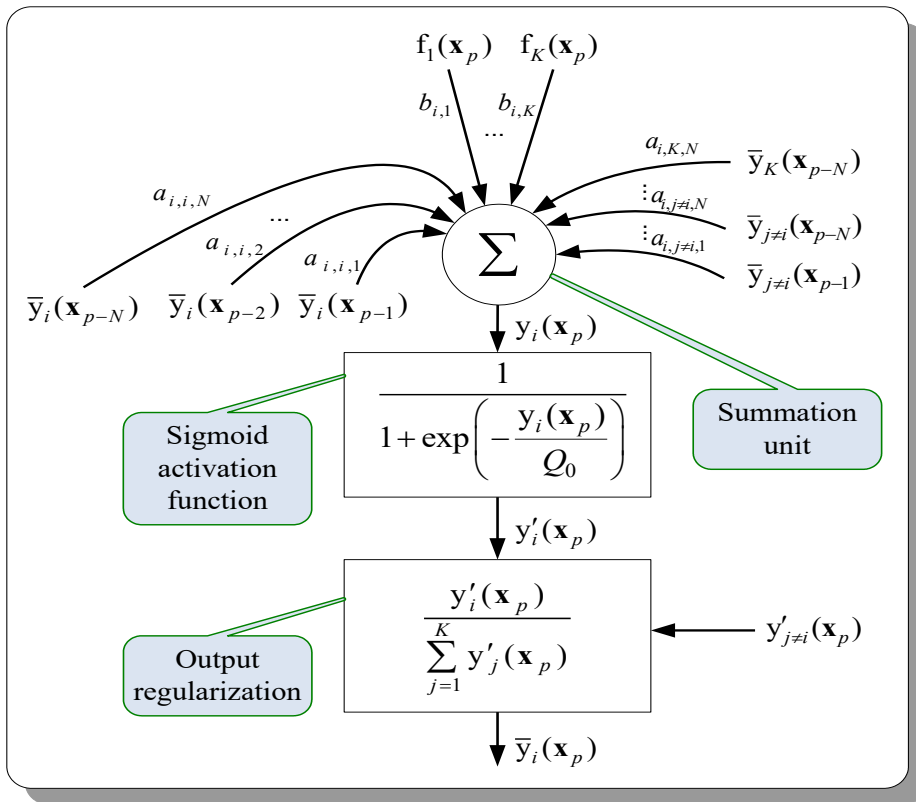
# Locally Recurrent Probabilistic Neural Networks



$i$ - неврон на рекурентния слой на LRPNN



връзки



Вътрешна структура

# Locally Recurrent PNN

$$f_i(\mathbf{x}_p) = \frac{1}{(2\pi)^{d/2} \sigma_i^d} \cdot \frac{1}{M_i} \sum_{j=1}^{M_i} \exp\left(-\frac{1}{2\sigma_i^2} (\mathbf{x}_p - \mathbf{x}_{ij})^T (\mathbf{x}_p - \mathbf{x}_{ij})\right)$$

$$y_i(\mathbf{x}_p) = \left[ b_{i,i} f_i(\mathbf{x}_p) - \sum_{\substack{k=1 \\ i \neq k}}^K b_{i,k} f_k(\mathbf{x}_p) \right] + \sum_{t=1}^N \left[ a_{i,i,t} \bar{y}_i(\mathbf{x}_{p-t}) - \sum_{\substack{k=1 \\ i \neq k}}^K a_{i,k,t} \bar{y}_k(\mathbf{x}_{p-t}) \right]$$

$$\bar{y}_i(\mathbf{x}_p) = \frac{\text{sgm}(y_i(\mathbf{x}_p))}{\sum_{j=1}^K \text{sgm}(y_j(\mathbf{x}_p))}$$

$$D(\mathbf{x}_p) = \underset{i}{\operatorname{argmax}} \{ h_i c_i \bar{y}_i(\mathbf{x}_p) \}$$

$$P(k_i | \mathbf{x}_p) = \frac{h_i \bar{y}_i(\mathbf{x}_p)}{\sum_{j=1}^K h_j \bar{y}_j(\mathbf{x}_p)}$$

# Locally Recurrent Probabilistic Neural Networks



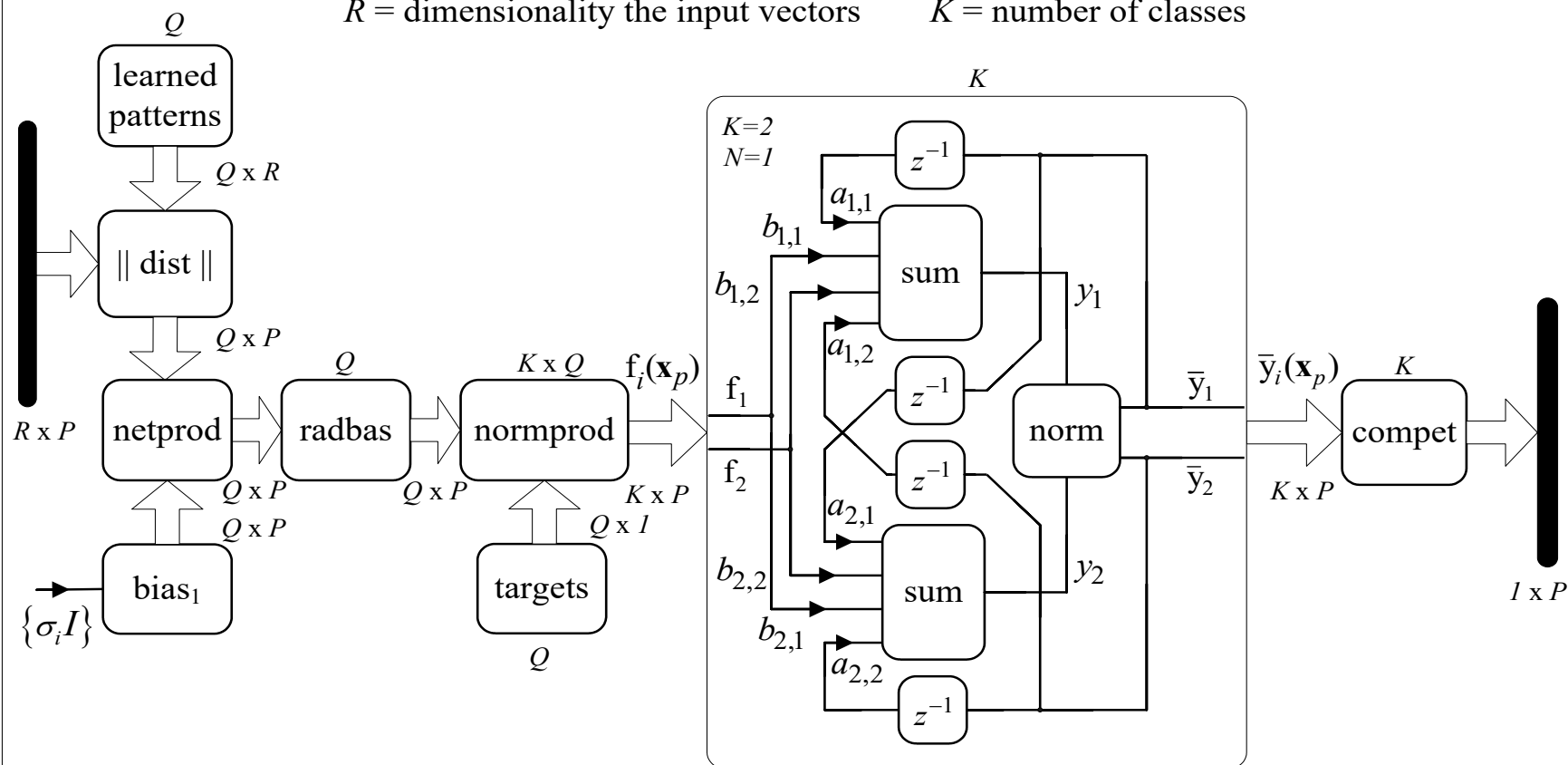
Architecture of the Locally Recurrent Probabilistic Neural Network

$P$  = number of input vectors

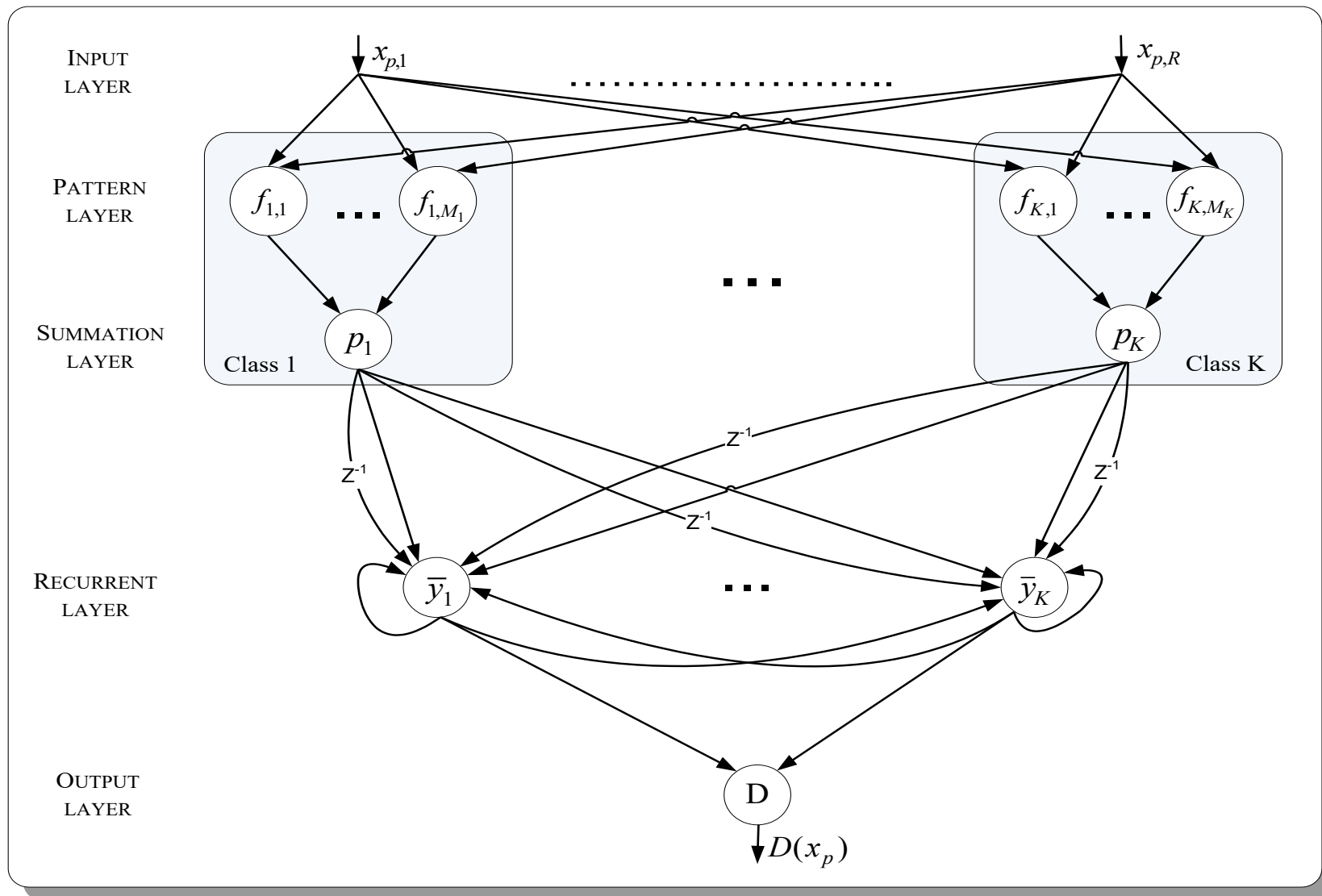
$Q$  = number of pattern units

$R$  = dimensionality the input vectors

$K$  = number of classes



# Generalized LR PNN



# Generalized LR PNN

$$f_i(\mathbf{x}_p) = \frac{1}{(2\pi)^{d/2} \sigma_i^d} \cdot \frac{1}{M_i} \sum_{j=1}^{M_i} \exp \left( -\frac{1}{2\sigma_i^2} (\mathbf{x}_p - \mathbf{x}_{ij})^T (\mathbf{x}_p - \mathbf{x}_{ij}) \right),$$

$$y_i(\mathbf{x}_p) = \sum_{t=0}^L \sum_{k=1}^K b_{i,k,t} f_k(\mathbf{x}_{p-t}) + \sum_{t=1}^N \sum_{k=1}^K a_{i,k,t} \bar{y}_k(\mathbf{x}_{p-t})$$

$$\bar{y}_i(\mathbf{x}_p) = \frac{\text{sgm}(y_i(\mathbf{x}_p))}{\sum_{j=1}^K \text{sgm}(y_j(\mathbf{x}_p))}$$

$$D(\mathbf{x}_p) = \underset{i}{\operatorname{argmax}} \{ h_i c_i \bar{y}_i(\mathbf{x}_p) \}$$

$$P(k_i | \mathbf{x}_p) = \frac{h_i \bar{y}_i(\mathbf{x}_p)}{\sum_{j=1}^K h_j \bar{y}_j(\mathbf{x}_p)}$$

# Generalized LR PNN

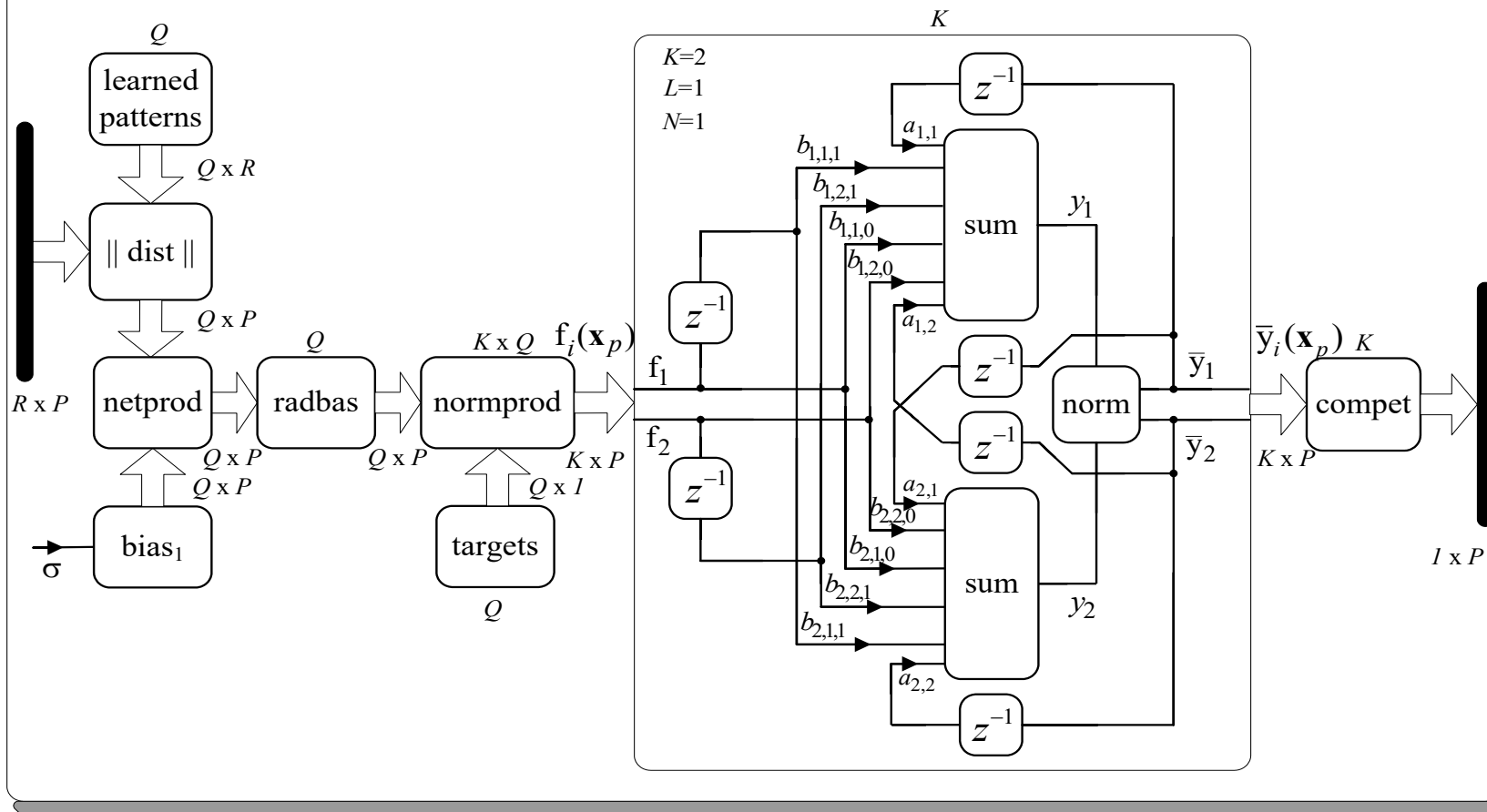
## Architecture of the Generalized Locally Recurrent Probabilistic Neural Network

$P$  = number of input vectors

$R$  = dimensionality the input vectors

$Q$  = number of pattern units

$K$  = number of classes



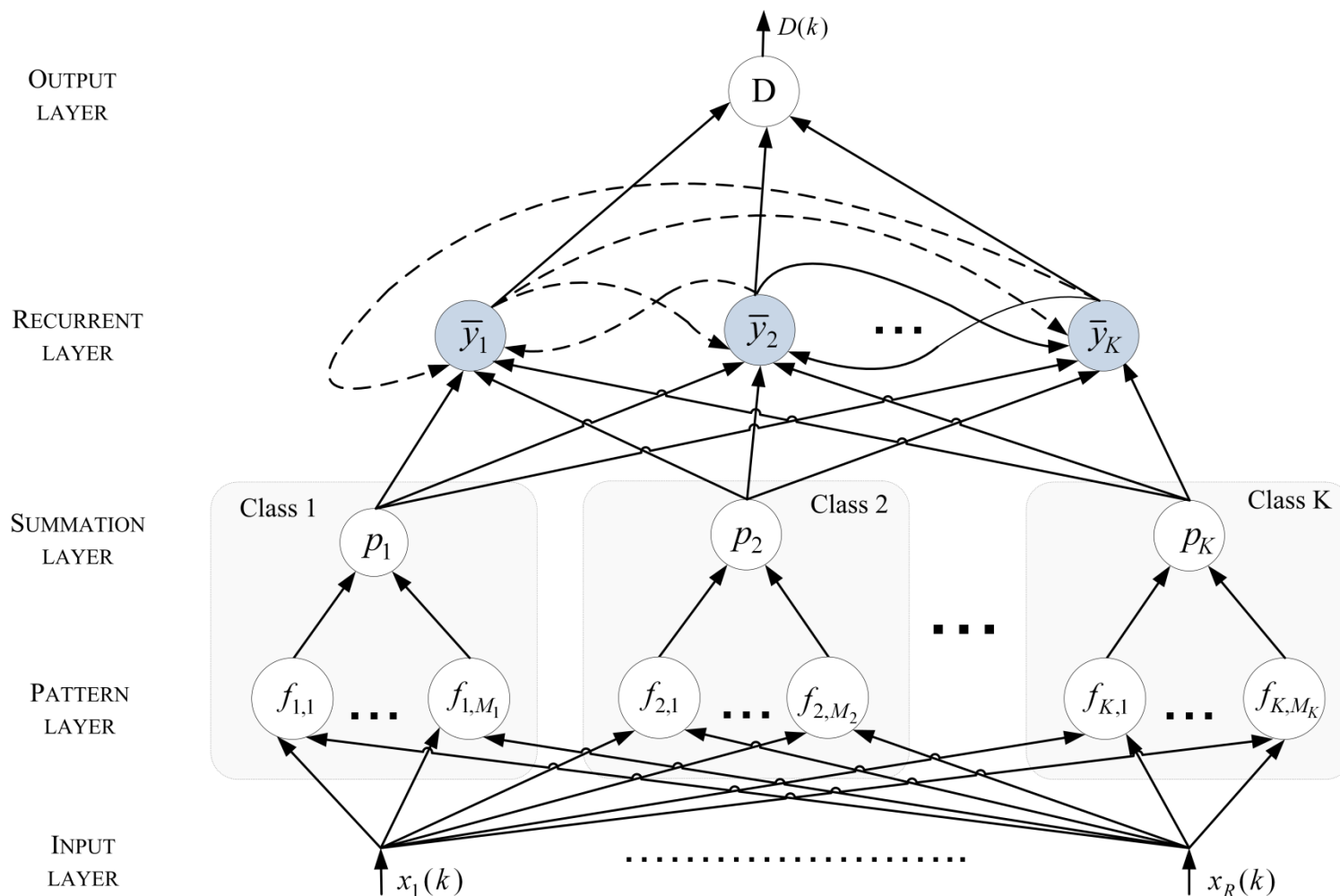


# Generalized LR PNN

The EER and  $DCF_{\text{opt}}$  depending on the architecture

|         | Arch.<br># $w^*$   | FIR<br>4 | IIR<br>6 | DRNN<br>10 | LR PNN<br>8 | GLR PNN<br>12 |
|---------|--------------------|----------|----------|------------|-------------|---------------|
| Males   | EER (%)            | 4.45     | 4.17     | 4.02       | 3.98        | 3.89          |
|         | $DCF_{\text{opt}}$ | 0.307    | 0.268    | 0.249      | 0.241       | 0.230         |
| Females | EER (%)            | 5.32     | 5.22     | 5.00       | 4.91        | 4.70          |
|         | $DCF_{\text{opt}}$ | 0.447    | 0.424    | 0.421      | 0.389       | 0.378         |

# Partially Connected Locally Recurrent Probabilistic Neural Networks



# Partially Connected Locally Recurrent Probabilistic Neural Networks



$$p_j(\mathbf{x}(k) | j) = f_j(\mathbf{x}_k) = \frac{1}{(2\pi)^{d/2} \sigma_j^d} \cdot \frac{1}{M_j} \sum_{i=1}^{M_j} \exp\left(-\frac{1}{2\sigma_j^2} (\mathbf{x}_k - \mathbf{x}_{ij})^T (\mathbf{x}_k - \mathbf{x}_{ij})\right), j = 1, 2, \dots, K,$$

$$y_j(\mathbf{x}_k) = b_{jj} f_j(\mathbf{x}_k) - \sum_{\substack{i=1 \\ i \neq j}}^K b_{ij} f_j(\mathbf{x}_k) + \sum_{t=1}^N (a_{jjt} \bar{y}_j(\mathbf{x}_{k-t}) - \sum_{\substack{i=1 \\ i \neq j}}^K a_{ijt} \bar{y}_j(\mathbf{x}_{k-t})), j = 1, 2, \dots, K,$$

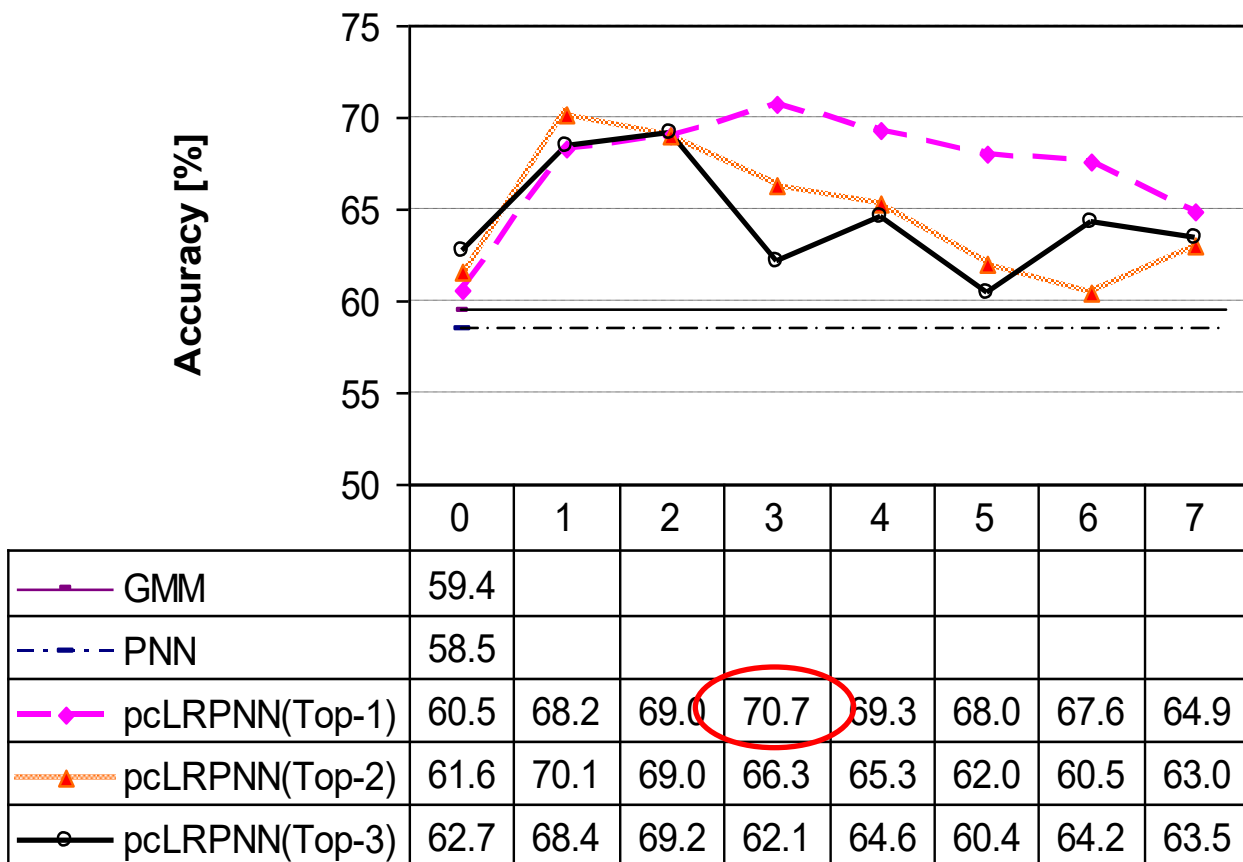
$$\bar{y}_j(\mathbf{x}_k) = \frac{\text{sgm}(y_j(\mathbf{x}_k))}{\sum_{i=1}^K \text{sgm}(y_i(\mathbf{x}_k))}, j = 1, 2, \dots, K,$$

$$D(\mathbf{x}_k) = \underset{j}{\operatorname{argmax}} \{h_j c_j \bar{y}_j(\mathbf{x}_k)\}, j = 1, 2, \dots, K,$$

$$E(\mathbf{w}) = \sum_{i=1}^K m_i \left(1 - P(\mathbf{X}_{k_i} | k_i)\right) P(k_i) + \frac{1}{K-1} \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K m_j P(\mathbf{X}_{k_i} | k_j) P(k_j)$$

# pcLRPNN

## Разпознаване на емоционална реч



# Класификатори

